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CO1- AT-1

CSA0904

1) Student Management System

```
class student
Private name, roll No, marks
  Method acceptDetails ()
    Input Name, rollNo, marks
  END Method
  Method calculateAverage ()
      average = Marks
      Return average
    END Method
  Method display Result()
      If marks >=50 then
          Print "Pass"
      Else
          Print "fail"
      END if
      END Method
    END class
    Create students
    CALL S. acceptDetails ()
    CALL S. calculate Average()
    CALL S. displayResult()
    STOP
```

2) Library Book Management

```
Class Book {
    Private string title, author, isbn;
    Private boolean is Available = true
        Method issue Book() {
        If (is Available = = true) {
         is available = false
         Print "Book Issued"
         }
         Else
          Print "Book Not Available"
        }
        Method return Book()
        {
         isAvailable = true
          Print "Book" Returned"
        }
        Method displayBook() {
        Print title, author, ISBN, isAvailable
         }
        }
        Main() {
        Create Book b1, b2, b3
         b1.displayBook()
         b1.issue Book()
          b1.return Book ()
        }
```

3) Employee salary calculation

```
class Employee {
  Private empId, name, basicPay
   calculate Gross Salary()
   calculate Net salary()
   display Pay slip()
  }
  Main() {
   Create employee e
    e.displayplaySlip()
    }
```

4) Shape Area calculator

```
class shape {
  calculate Area() }
  class circle Extends shape
  class Rectangle Extend shape
  class Triangle extend shape
   Main() {
   shape s
    s= new circle()
    s.calculateArea()
    s= new Rectangle()
```

```
s.calculateArea()

s= new Triangle()
s.calculate Area()
}
```

5.) vehicle Registration system

```
class vehicle {
RegNo ,owner, fuel-type
}
class Twowheeler extends vehicle {
engine capacity
}
class fourwheeler extends vehicle {
seating capacity
}
Main() {
Create Twowheeler t
create Four wheeler f
}
```

6) Bank Account Hierarchy

```
class BankAccount{
accNo, balance , holder withdraw()
}
```

```
class savingsAccount Extends BankAccount {
    withdraw()
   }
  class currentAccount Extends
 BankAccount {
        withdraw()
    }
   Main()
   {
   Create savingsAccount s
   Create currentAccount c
    s.withdraw()
    c.withdraw()
```

7) Hospital Patient Record system

```
 class Patient {
 Private diagnosis, Prescription, generateBill()
 }
class InPatient Extends Patient {
     generate Bill()
     }
class outPatient Extends Patients {
     generate Bill()
```

```
Main() {
    Create Inpatient I
    Create Outpatient O
     I.generate Bill()
     O.generate Bill()
```

6) E-commerce Product Catalog

```
class Product {
    apply Discount()
}
class Electronics extends product {
    apply Discount()
}
class clothing extends Product {
    apply Discount()
}
Main()
{
    create Electronics e
    create clothing c
    create Grocery g
    e.applyDiscount()
    c.applyDiscount()
    g.apply Discount()
}
```

9) University Staff Hierarchy

```
class staff {
    CalculateAllowance()
}
class Professor extends Staff {
    calculate Allowance()
}
Class Lecturer extends staff {
    calculate Allowance()
}
Main() {
    create professor P
    create lecturer l
        P. calculate Allowance()
        l. calculate Allowance()
}
```

10)

```
class smart Device {
    turnon()
    turn off()
}
class Light extends smart Device {
    turnon()
}
```

```
class fan extends smartDevice {

    turnOn()

  }

Main() {

    Create light l
    create fan f

    l. turnOn()
    f.turnOn()

    }
```